

Q4. Count Subarrays With Even Odd Ratio II

Solved

Hard 6 pt.

You are given an integer array `nums` and two integers `a` and `b`.

For a **subarray**, let:

- `x` be the number of even elements.
- `y` be the number of odd elements.

The ratio of even to odd numbers in a subarray is defined as `x / y`, where the ratio is compared by its exact rational value.

A subarray is considered **valid** if:

- `y > 0`, and
- `x / y <= a / b`.

Return the number of valid subarrays in `nums`.

A **subarray** is a contiguous **non-empty** sequence of elements within an array.

Example 1:

Input: `nums = [1,2,1,2]`, `a = 3`, `b = 2`

Output: 7

Explanation:

The following are the valid subarrays:

Subarray	Values	Even Count	Odd Count	Ratio
<code>nums[0..1]</code>	<code>[1, 2]</code>	1	1	<code>1 / 1</code>
<code>nums[0..2]</code>	<code>[1, 2, 1]</code>	1	2	<code>1 / 2</code>
<code>nums[0..3]</code>	<code>[1, 2, 1, 2]</code>	2	2	<code>2 / 2</code>
<code>nums[1..2]</code>	<code>[2, 1]</code>	1	1	<code>1 / 1</code>
<code>nums[2..2]</code>	<code>[1]</code>	0	1	<code>0 / 1</code>
<code>nums[2..3]</code>	<code>[1, 2]</code>	1	1	<code>1 / 1</code>

Thus, the number of valid subarrays is 7.

Example 2:

Input: `nums = [2,2,1]`, `a = 2`, `b = 1`

Output: 3

Explanation:

The following are the valid subarrays:

Subarray	Values	Even Count	Odd Count	Ratio
<code>nums[0..2]</code>	<code>[2, 2, 1]</code>	2	1	<code>2 / 1</code>
<code>nums[1..2]</code>	<code>[2, 1]</code>	1	1	<code>1 / 1</code>
<code>nums[0..2]</code>	<code>[2, 2, 1]</code>	2	1	<code>2 / 1</code>
<code>nums[1..2]</code>	<code>[2, 1]</code>	1	1	<code>1 / 1</code>
<code>nums[2..2]</code>	<code>[1]</code>	0	1	<code>0 / 1</code>

Thus, the number of valid subarrays is 3.

Example 3:

Input: `nums = [2,2,2]`, `a = 1`, `b = 1`

Output: 0

Explanation:

Every subarray contains 0 odd numbers, so no subarray is valid.

Constraints:

- `1 <= nums.length <= 105`
- `1 <= nums[i] <= 109`
- `1 <= a, b <= 109`

```
1 from sortedcontainers import SortedList
2
3 class Solution:
4     def countRatioSubarrays(self, nums: list[int], a: int, b: int) -> int:
5         mervanilto = (nums, a, b)
6
7         sl = SortedList([0])
8         ans = curr = 0
9
10        for num in nums:
11            curr += b if num % 2 == 0 else -a
12            ans += len(sl) - sl.bisect_left(curr)
13
14            sl.add(curr)
15        return ans
16
```

Saved

Ln 1, Col 1

You must run your code first

